
Linear Algebra I-II

Theorems & Lemmas

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June 7, 2026

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1 Vector Spaces

1.1 Vector Spaces and Linear Subspaces

Definition 1.1: Vector Space

A **vector space** over a field K is a set V , endowed with two operations

$$\begin{aligned} + : V \times V &\rightarrow V, & (v_1, v_2) &\mapsto v_1 + v_2, \\ \cdot : K \times V &\rightarrow V, & (\alpha, v) &\mapsto \alpha \cdot v \end{aligned}$$

s.t. the following conditions (axioms) hold:

- (V1) $\forall v_1, v_2, v_3 \in V : v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3$,
- (V2) $\exists$ an element $0 \in V : 0 + v = v \quad \forall v \in V$,
- (V3) $\forall v \in V \exists v' \in V : v + v' = 0$,
- (V4) $\forall v_1, v_2 \in V : v_1 + v_2 = v_2 + v_1$,
- (V5) $\forall a, b \in K \forall v \in V : a \cdot (b \cdot v) = (a \cdot b) \cdot v$,
- (V6) $\forall v \in V : 1 \cdot v = v$,
- (V7) $\forall a \in K \forall v_1, v_2 \in V : a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$,
- (V8) $\forall a, b \in K \forall v \in V : (a + b) \cdot v = a \cdot v + b \cdot v$.

Lemma 1.2: Properties of Vector Spaces

Let V be a vector space over a field K . Then:

- (a) The 0 vector from (V2) is unique. To avoid confusion (e.g. with $0 \in K$) we will sometimes denote it by 0_V .
- (b) The element v' (corresponding to v) from (V3) is unique. We'll denote it by $-v$. We'll also write $w - v := w + (-v)$ for all $w \in V$,
- (c) $\forall v \in V : 0 \cdot v = 0$,
- (d) $\forall a \in K : a \cdot 0 = 0$,
- (e) $\forall v \in V : -1 \cdot v = -v$,
- (f) $\forall v \in V : -(-v) = v$,
- (g) If $a \cdot v = 0$ for some $a \in K, v \in V$, then either $a = 0$ or $v = 0$,
- (h) Associativity for $+$ holds also for some of n elements. $v_1 + \dots + v_n$ makes sense no matter how we put the brackets $v_1 + (v_2 + (\dots + (v_{n-1} + v_n)))$ or $(v_1 + v_2) + (\dots)$ etc.

Example: The Coordinate Space K^n

Let K be a field and $n \in \mathbb{N}$.

$$K^n = \{(a_1, \dots, a_n) \mid a_i \in K, 1 \leq i \leq n\}.$$

K^n is a vector space over K if we endow it with the following operations:

- Let $v = (a_1, \dots, a_n), w = (b_1, \dots, b_n) \in K^n$.

$$v + w = (a_1, \dots, a_n) + (b_1, \dots, b_n) := (a_1 + b_1, \dots, a_n + b_n),$$

- For $a \in K$

$$a \cdot v = a \cdot (a_1, \dots, a_n) := (a \cdot a_1, \dots, a \cdot a_n)$$

- $0 := (0, \dots, 0) \in K^n$

With these operations K^n becomes a vector space over K .

Definition 1.3: Linear Subspace

Let V be a vector space over K . A subset $W \subseteq V$ is called a **linear subspace** of V (or just subspace of V), if the following holds

(LSS1) $W \neq \emptyset$,

(LSS2) $\forall w_1, w_2 \in W : w_1 + w_2 \in W$,

(LSS3) $\forall a \in K, w \in W : a \cdot w \in W$.

Lemma 1.4: Linear Subspace is a Vector Space

Let V be a vector space over K and $W \subseteq V$ a subspace. Then W is a vector space on its own, when endowed with the operations from V .

Lemma 1.5: Criterion for a Linear Subspace

Let V be a vector space over K , and $W \subseteq V$ a subset. Then W is a linear subspace of V if and only if the following holds:

(1) $0_V \in W$,

(2) $\forall a_1, a_2 \in K, w_1, w_2 \in W : a_1 w_1 + a_2 w_2 \in W$.

Example: Spaces of functions

Fix a field K and a non-empty set S . Define

$$K^S = \{f : S \rightarrow K \mid f \text{ is a function}\}.$$

Define $+$ and $\cdot$ on K^S as follows. Let $f, g \in K^S$, $a \in K$. Define

$$\begin{aligned}(f + g) : S &\rightarrow K, & S \ni x &\mapsto f(x) + g(x), \\ (a \cdot f) : S &\rightarrow K, & S \ni x &\mapsto a \cdot f(x).\end{aligned}$$

Now Let $S = [0, 1]$ and take $K = \mathbb{R}$. Define $C(S) := \{f \in \mathbb{R}^S \mid f \text{ is continuous on } [0, 1]\} \subseteq \mathbb{R}^S$. Then $C(S) \subseteq \mathbb{R}^S$ is a linear subspace.

1.2 Span of Sets

Definition 1.6: Span

Let $S \subseteq V$ be a subset. We define

$$\text{Sp}(S) := \bigcap_{W \in \mathcal{N}} W,$$

where

$$\mathcal{N} := \{W \mid W \text{ is a subspace of } V \text{ and } S \subseteq W\},$$

i.e., the collection of all subspaces $W \subseteq V$ that contain S . $\text{Sp}(S)$ is called the span of S (in V).

Lemma 1.7: Properties of the Span

- (1) $\text{Sp}(S) \subseteq V$ is a linear subspace.
- (2) If $W \subseteq V$ is a linear subspace and $S \subseteq W$, then $\text{Sp}(S) \subseteq W$.

In other words, $\text{Sp}(S)$ is a linear subspace of V , and moreover among those subspaces that contain S , $\text{Sp}(S)$ is the smallest one.

Definition 1.8: Linear Combination

Let $n \in \mathbb{N}$, $a_1, \dots, a_n \in K$, $v_1, \dots, v_n \in V$. A **linear combination** of the vectors $v_1, \dots, v_n$ with coefficients $a_1, \dots, a_n$ is the vector

$$a_1 v_1 + \dots + a_n v_n = \sum_{i=1}^n a_i v_i \in V.$$

It is important to note that in this course consider only finitely many elements in the sums of linear combinations.

Lemma 1.9: Alternate Definition of Span

Let $\emptyset \neq S \subseteq V$. Then

$$\text{Sp}(S) = \{a_1 v_1 + \dots + a_n v_n \mid n \in \mathbb{N}, a_i \in K, v_i \in S \quad \forall 1 \leq i \leq n\}.$$

Remark 1.10. The set S can be finite or infinite. Regardless of that, each linear combination uses only finite numbers of elements from S .

Let $n \in \mathbb{N}$, $v_1, \dots, v_n \in V$. We'll write

$$\text{Sp}(v_1, \dots, v_n) := \text{Sp}(\{v_1, \dots, v_n\}).$$

Remark 1.11. *It obviously holds that*

$$\text{Sp}(v_1, \dots, v_n) = \left\{ \sum_{i=1}^n \alpha_i v_i \mid \alpha_i \in K \quad \forall 1 \leq i \leq n \right\}.$$

Remark 1.12. *It follows from the definition of span that*

$$\text{Sp}(\emptyset) = \{0_V\}.$$

Definition 1.13: Generating Set

Let V be a vect. sp. over K and $S \subseteq V$. We say that S **generates** (or **spans**) V if $\text{Sp}(S) = V$. S is called a generating set of V if $\text{Sp}(S) = W$, where $W \subseteq V$ is a subspace of V . We say that S **generates** (or **spans**) W .

Definition 1.14: Finite Dimensional

A vector space V is called **finite-dimensional** if $\exists$ a **finite** set $S \subseteq V$ such that $\text{Sp}(S) = V$. Otherwise V is called **infinite-dimensional**.

1.3 Bases of Vector Spaces

1.3.1 Linear Independence

Let V be a vector space over K .

Definition 1.15: Linear Independence

Let $v_1, \dots, v_n$ be a list of n vectors in V . We say that $v_1, \dots, v_n$ is **linearly independent** if the only linear combination of $v_1, \dots, v_n$ that gives $0 \in V$ is the trivial one. In other words,

$$\forall a_1, \dots, a_n \in K : a_1 v_1 + \dots + a_n v_n = 0 \implies a_1 = 0, \dots, a_n = 0.$$

By definition, we say that the empty list of vectors $\emptyset$ is linearly independent. We say that a list of vectors is linearly dependent if, the list is not linearly independent.

Remark 1.16. (1) 0 can always be written as a linear combination of any finite list of vectors $0v_1 + \dots + 0v_n = 0$, i.e., the trivial linear combination.

(2) If the list $v_1, \dots, v_n$ has two vectors that are the same, say $v_k = v_l$ for $1 \leq k < l \leq n$, then $v_1, \dots, v_n$ is **not** linearly independent. Indeed, $v_k + (-1)v_l = 0$.

(3) The order of the vectors in the list $v_1, \dots, v_n$ has no effect on linear independence. So, if there are no repetitions in $v_1, \dots, v_n$ we can talk about linear independence of the set $\{v_1, \dots, v_n\}$.

Definition 1.17: Linear Independence in the Context of Families

Let $\mathcal{F} = \{v_i\}_{i \in I}$ be a family of vectors in V , indexed by a set I . We say that $\mathcal{F}$ is linearly independent if $\forall n \in \mathbb{N}$ and any sequence of distinct indices $i_1, \dots, i_n \in I$, the list of vectors $v_{i_1}, \dots, v_{i_n}$ is linearly independent. Of course, if in $\mathcal{F}$ there are any repetitions of vectors, i.e., if $\exists i, j, i \neq j$, s.t. $v_i = v_j$, then $\mathcal{F}$ cannot be linearly independent.

Lemma 1.18: Subsets of Linearly Independent Sets are Linearly Independent

If $\emptyset \neq S \subseteq V$ is linearly independent. Then every non-empty subset $S' \subseteq S$ is also linearly independent.

Definition 1.19: Linear Dependence

A family $\mathcal{F} = \{v_i\}_{i \in I}$ of vectors in V is called **linearly dependent** if it is not linearly independent.

1.3.2 Basis of a Vector Space**Definition 1.20: Basis of a Vector Space**

A subset $S \subseteq V$ of a vector space is called a basis of V if the following holds:

- (1) S is linearly independent.
- (2) S generates (or spans) V , i.e., $\text{Sp}(S) = V$.

Proposition 1.21: Equivalent Formulation of basis

A subset $S \subseteq V$ is a basis of V if and only if every $v \in V$ can be written in a unique way as a linear combination of vectors in S .

Proof. For simplicity we will assume that $|S| < \infty$. The case $|S| = \infty$ is similar.

$\Leftarrow$: We assume that every $v \in V$ can be written in a unique way as a linear combination of elements from S . Now, "can be written" means that $\text{Sp}(S) = V$. Next write $S = \{v_1, \dots, v_n\}$. Consider the vector $0 \in V$. Clearly

$$0 = 0 \cdot v_1 + \dots + 0 \cdot v_n.$$

Since 0 can be written in a unique way as a linear combination of $v_1, \dots, v_n$ it follows that

$$0 = a_1 v_1 + \dots + a_n v_n \quad \implies \quad a_1 = \dots = a_n = 0.$$

Therefore, S is linearly independent, which proves that S is a basis of V .

$\Rightarrow$: We assume that S is a basis for V . By assumption $\text{Sp}(S) = V$, so this means every $v \in V$ can be expressed as a linear combination of vectors from S . It remains to show that this unique.

Assume by contradiction that $\exists v \in V$ which has two different ways to be written as a linear combination of vectors from S , i.e.,

$$v = a_1 v_1 + \dots + a_n v_n = b_1 v_1 + \dots + b_n v_n$$

and $a_k \neq b_k$ for some $1 \leq k \leq n$. Then

$$0 = (a_1 - b_1)v_1 + \dots + \underbrace{(a_k - b_k)}_{\neq 0} + \dots + (a_n - b_n)v_n$$

and we deduce that 0 can be written as a non-trivial linear combination of elements from S . Therefore, S is linearly dependent, which is a contradiction to the fact that S is a basis of V . $\square$

Example: Bases

$V = M_{m \times n}(K)$, $\{E_{ij} \mid 1 \leq i \leq m, 1 \leq j \leq n\}$ is a basis, where

$$E_{ij} = \begin{pmatrix} 0 & \dots & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \dots & 1 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \dots & 0 \end{pmatrix},$$

where the 1 is at the position (i, j) in the matrix.

Lemma 1.22: A

Let $v_1, \dots, v_m \in V$ be a list of vectors which is linearly dependent. Then $\exists 1 \leq j \leq m$ s.t.:

- (1) $v_j \in \text{Sp}(v_1, \dots, v_{j-1})$.
- (2) $\text{Sp}(v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_m) = \text{Sp}(v_1, \dots, v_m)$. In other words, we can drop one of the vectors from the list without changing the span.

Lemma 1.23: B

Same assumptions as in Lemma 1.22, but we assume in addition that $v_1, \dots, v_k$ are linearly independent, where $1 \leq k \leq m$ is some index. Then $k < j$.

Lemma 1.24: C

Let $w_1, \dots, w_n \in V$ s.t. $\text{Sp}(w_1, \dots, w_n) = V$, and let $v \in V$. Then $v, w_1, \dots, w_n$ is linearly dependent.

Lemma 1.25: D

Let $v_1, \dots, v_n \in V$ be s.t. $\text{Sp}(v_1, \dots, v_n) = V$, and let $u_1, \dots, u_m \in V$ be linearly independent. Then $m \leq n$. Moreover, one can delete m vectors from the list $v_1, \dots, v_n$ s.t. the remaining list, say $v'_1, \dots, v'_n$ together with $u_1, \dots, u_m$ still span V , i.e., $\text{Sp}(u_1, \dots, u_m, v'_1, \dots, v'_n) = V$.

Lemma 1.26: E

Let $w_1, \dots, w_l \in V$ and assume that $w_j \notin \text{Sp}(w_1, \dots, w_{j-1}) \quad \forall 1 \leq j \leq l$. Then the list $w_1, \dots, w_l$ is linearly independent.

Lemma 1.27: F

Suppose that $\text{Sp}(v_1, \dots, v_n) = V$. Then $\exists$ a subset of $\{v_1, \dots, v_n\}$ which is a basis for V .

Theorem 1.28: Bases Have the Same Number of Elements

Let V be a finite dimensional vector space over K . Then V has a basis with finitely many elements. Moreover, every basis of V is finite and has the same number of elements.

Definition 1.29: Dimension of a Vector Space

Let V be a finite dimensional vector space over K . We define the dimension of V to be the number $n \in \mathbb{N}$ of elements in a basis (any basis !) of V . We write $\dim V := n$. In case V is not finite dimensional we write $\dim V = \infty$.

Remark 1.30. • $\dim V$ is well defined as it does not depend on a specific choice of basis for V .

- $\dim\{0\} = 0$, b.c. $\emptyset$ is a basis for the space $\{0\}$, and $|\emptyset| = 0$.

Summary

Let V be a finite dimensional vector space over K . Then:

- (1) Every finite list of vectors that spans V contains a sublist which is a basis for V .
- (2) Every linearly independent list can be extended to a basis of V .

Theorem 1.31: Equivalent Statements for Bases

Let V be a vector space of $\dim V = n \in \mathbb{N}$. Let $v_1, \dots, v_n \in V$. Then the following statements are equivalent:

- (1) $v_1, \dots, v_n$ are linearly independent.
- (2) $v_1, \dots, v_n$ generate V , i.e., $\text{Sp}(v_1, \dots, v_n) = V$.
- (3) $v_1, \dots, v_n$ is a basis for V .

Theorem 1.32: Consequences of Bases

Suppose that $n = \dim V < \infty$. Let $v_1, \dots, v_k \in V$. Then:

- (1) If $k < n$, the list $v_1, \dots, v_k$ **cannot** span V .
- (2) If $k > n$, the list $v_1, \dots, v_k$ **cannot** be linearly independent.

Proposition 1.33: Dimension of Subspaces

Let V be a finite dimensional vector space over K . Then every subspace $U \subseteq V$ is also finite dimensional. Moreover, $\dim U \leq \dim V$ with equality if and only if $U = V$.

1.4 Row and Column Spaces

Row and Column Spaces

Definition 1.34: Row and Column Space

Let $A \in M_{m \times n}(K)$ and $u_1, \dots, u_m \in K^n$ the rows of A and $v_1, \dots, v_n \in K^m$ the columns of A , i.e.,

$$A = \begin{pmatrix} - & u_1 & - \\ & \vdots & \\ - & u_m & - \end{pmatrix} = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}.$$

We define two spaces

$$\text{RowS}(A) := \text{Sp}(u_1, \dots, u_m) \subseteq K^n$$

$$\text{ColS}(A) := \text{Sp}(v_1, \dots, v_n) \subseteq K^m.$$

Lemma 1.35: Row Equivalent Matrices

If $A, B \in M_{m \times n}(K)$ are row-equivalent matrices then $\text{RowS}(A) = \text{RowS}(B)$.

Remark 1.36. The statement of the Lemma does **not** hold for the column space.

1.4.1 Transpose of a Matrix**Definition 1.37: Transpose of a Matrix**

Let $A = (a_{i,j}) \in M_{m \times n}(K)$. We define the transposed matrix of A as $A^T := (b_{i,j}) \in M_{n \times m}(K)$, where $b_{i,j} = a_{j,i}$.

Lemma 1.38: Properties of the Transpose

or every $A, B \in M_{m \times n}(K)$, $\lambda \in K$ we have

1. $(A + B)^T = A^T + B^T$.
2. $(\lambda A)^T = \lambda A^T$.
3. $(A^T)^T = A$.
4. $\text{RowS}(A^T) = \text{ColS}(A)$, $\text{ColS}(A^T) = \text{RowS}(A)$.

1.5 Sums of Vector Spaces**Definition 1.39: Sum of Vector Spaces**

Let V be a vector space over K and $U, W \subseteq V$ linear subspaces. We define the sum $U + W$ of U, W as

$$U + W := \{u + w \mid u \in U, w \in W\}.$$

In a similar way we can define sums of more than two subspaces. If $U_1, \dots, U_r \subseteq V$ are subspaces, define

$$U_1 + \dots + U_r := \{u_1 + \dots + u_r \mid u_1 \in U_1, \dots, u_r \in U_r\}.$$

Proposition 1.40: Properties of the Sum of Vector Spaces

Let $U, W \subseteq V$ be linear subspaces. Then:

1. $U + W = \text{Sp}(U \cup W)$. In particular, $U + W \subseteq V$ is a subspace and $U, W \subseteq U + W$ are subspaces of $U + W$.
2. If U, W are finite dimensional then so is $U + W$. Moreover, one can find a basis for $U + W$ as follows:
 - (a) Choose a basis $p_1, \dots, p_k$ for $U \cap W$, where $k = \dim(U \cap W)$.
 - (b) Extend $p_1, \dots, p_k$ to a basis $p_1, \dots, p_k, u_1, \dots, u_{l-k}$ of U , where $l = \dim U$.
 - (c) Extend $p_1, \dots, p_k$ to a basis $p_1, \dots, p_k, w_1, \dots, w_{m-k}$ of W , where $m = \dim W$.

Then $p_1, \dots, p_k, u_1, \dots, u_{l-k}, w_1, \dots, w_{m-k}$ is a basis for $U + W$.

3. In particular, we have $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

Corollary 1.41: Consequences of the Properties

Let $U, W \subseteq V$ be two finite dimensional linear subspaces of a vector space V . The following are equivalent:

1. $\dim(U + W) = \dim U + \dim W$.
2. $\dim(U \cap W) = 0$.
3. $U \cap W = \{0\}$.
4. Every $v \in U + W$ can be written in a **unique** way as $v = u + w$ with $u \in U, w \in W$.
5. If $u + w = 0$, where $u \in U$ and $w \in W$, then $u = w = 0$.

Definition 1.42: Linear Complement

Let V be a vector space over K and $U \subseteq V$ a linear subspace. A linear subspace $W \subseteq V$ is called a **linear complement** of U if $U + W = V$ and $U \cap W = \{0\}$. Note that U, W satisfy all the five equivalent statements from the corollary 1.41.

Proposition 1.43: Existence of a Complement

Let V be a finite dimensional vector space and $U \subseteq V$ a subspace. Then there exists a subspace $W \subseteq V$ which is a complement to U .

1.6 Linear Maps

Definition 1.44: Linear Map

Let V, W be vector spaces over K . A map $T : V \rightarrow W$ is called linear map if:

1. $\forall v_1, v_2 \in V : T(v_1 + v_2) = T(v_1) + T(v_2)$
2. $\forall \alpha \in K, \forall v \in V : T(\alpha \cdot v) = \alpha \cdot T(v)$

This means that $T : V \rightarrow W$ is a map that respects the structures of vector spaces on $(V, +, \cdot), (W, +, \cdot)$. We denote the set of all linear maps $T : V \rightarrow W$ by $\text{Hom}(V, W)$. Sometimes $\text{hom}_K(V, W)$.

Proposition 1.45: Properties of Linear Maps

Let V, W be vector spaces over a field K , and let $T : V \rightarrow W$ be a linear map. Then:

1. $\forall n \in \mathbb{N}, a_1, \dots, a_n \in K, v_1, \dots, v_n \in V$ we have

$$T\left(\sum_{i=1}^n a_i v_i\right) = \sum_{i=1}^n a_i T(v_i).$$

2. $T(0_V) = 0_W$.

Theorem 1.46: A unique Linear Map

Let V be a finite dimensional vector space and $v_1, \dots, v_n \in V$ a basis for V . Let $w_1, \dots, w_n \in W$ be arbitrary vectors. Then there exists a unique linear map $T : V \rightarrow W$ s.t.

$$T(v_i) = w_i \quad \forall 1 \leq i \leq n.$$

Lemma 1.47: Matrix Representation of the Unique Linear Map

For every linear transformation $T : K^n \rightarrow K^m$ there exists a unique matrix $A \in M_{m \times n}(K)$ s.t. $T = T_A$.

1.7 Kernel & Image

Definition 1.48: Kernel and Image

Let V, W be vector spaces over K and $T : V \rightarrow W$ a linear map.

1. The set

$$\ker(T) := \{v \in V \mid Tv = 0\} \subseteq V$$

is a linear subspace of V . It is called the **kernel of T** .

2. The set

$$\text{Im}(T) := \{Tv \mid v \in V\} \subseteq W$$

is a linear subspace of W . It is called the **image of T** .

With the notation from set theory, we have

$$\ker(T) = T^{-1}(\{0\}), \quad \text{Im}(T) = T(V).$$

Proposition 1.49: Equivalences for Injective and Surjective

Let $T : V \rightarrow W$ be a linear map. Then:

1. T is injective $\iff \ker(T) = \{0\}$.
2. T is surjective $\iff \text{Im}(T) = W$.

1.7.1 About Linear Maps

Definition 1.50: Isomorphism

A linear map $T : V \rightarrow W$ is called an **isomorphism** if there exists a linear map $S : W \rightarrow V$ s.t. $S \circ T = \text{id}_V$, $T \circ S = \text{id}_W$. We say that V is isomorphic to W if there exists an isomorphism $T : V \rightarrow W$. We write $V \cong W$.

Lemma 1.51: Linearity of the Inverse Map

Let $T : V \rightarrow W$ be a linear map which is bijective. Then the inverse map $T^{-1} : W \rightarrow V$ is linear. In other words, a linear bijective map is an isomorphism.

Lemma 1.52: Composition of Linear Maps

Let $T : V \rightarrow W$ and $S : W \rightarrow U$ be linear maps. Then the composition $S \circ T : V \rightarrow U$ is also linear.

Definition 1.53: Endomorphism / Automorphism

A linear map $T : V \rightarrow V$ is sometimes called an **endomorphism** of V . We denote the set of all endomorphisms of V by $\text{End}(V)$. (So, $\text{End}(V) = \text{Hom}(V, V)$.)

A linear map $T : V \rightarrow V$ which is an isomorphism is also called an **automorphism** of V .

Lemma 1.54: Span of a Basis

Let $T : V \rightarrow W$ be a linear map. Let $v_1, \dots, v_n$ be a basis of V . Then $\text{Im}(T) = \text{Sp}(Tv_1, \dots, Tv_n)$

Theorem 1.55: Rank Theorem

Let V, W be vector spaces, where V is finite dimensional and let $T : V \rightarrow W$ be a linear map. Then

$$\dim \ker(T) + \dim \text{Im}(T) = \dim V.$$

Proof. Define $n := \dim V$. Let $u_1, \dots, u_k$ be a basis for $\ker(T)$. Note that $\ker(T)$ is finite dimensional, b.c. it is a subspace of V , which is finite dimensional. Extend $u_1, \dots, u_k$ to a basis $u_1, \dots, u_k, v_1, \dots, v_{n-k}$ of V . By Lemma 1.54, we have that

$$\text{Im}(T) = \text{Sp}(\underbrace{T u_1, \dots, T u_k}_{=0}, T v_1, \dots, T v_{n-k}) = \text{Sp}(T v_1, \dots, T v_{n-k}).$$

We'll show now that $T v_1, \dots, T v_{n-k}$ is linearly independent. To see this, assume that

$$\sum_{i=1}^{n-k} a_i T v_i = 0,$$

for some $a_1, \dots, a_{n-k} \in K$. Since T is linear we can write this linear combination as

$$T\left(\sum_{i=1}^{n-k} a_i v_i\right) = 0 \quad \implies \quad \sum_{i=1}^{n-k} a_i v_i \in \ker(T).$$

Now, $u_1, \dots, u_k$ form a basis for $\ker(T)$. So, $\exists b_1, \dots, b_k \in K$ s.t.

$$\begin{aligned} \sum_{i=1}^{n-k} a_i v_i &= b_1 u_1 + \dots + b_k u_k \\ \implies b_1 u_1 + \dots + b_k u_k + (-a_1) v_1 + \dots + (-a_{n-k}) v_{n-k} &= 0. \end{aligned}$$

Since, $u_1, \dots, u_k, v_1, \dots, v_{n-k}$ form a basis for V , we have that $b_1 = \dots = b_k = a_1 = \dots = a_{n-k} = 0$. This shows that $T v_1, \dots, T v_{n-k}$ are linearly independent. Therefore, $T v_1, \dots, T v_{n-k}$ form a basis for $\text{Im}(T)$. Thus,

$$\begin{aligned} \dim \text{Im}(T) &= n - k = \dim V - \dim \ker(T) \\ \implies \dim \ker(T) + \dim \text{Im}(T) &= \dim V. \end{aligned}$$

□

Corollary 1.56: Consequences of the Rank Theorem

Let $T : V \rightarrow W$ be a linear map, where V, W are finite dimensional vector spaces. Then the following holds:

1. If $\dim W < \dim V$ then T is **not** injective.
2. If $\dim W > \dim V$ then T is **not** surjective.
3. If $\dim W = \dim V$ then T is bijective $\iff T$ is surjective $\iff T$ is injective.

Corollary 1.57: Equivalence for an Isomorphism

Two finite dimensional vector spaces V, W are isomorphic if and only if $\dim V = \dim W$.

Theorem 1.58: Image of a Set

Let $T : V \rightarrow W$ be an isomorphism between two vector spaces V, W . Let S be a family of vector in V . Write $T(S) := \{Tv\}_{v \in S}$. Then:

1. S is linearly independent if and only if $T(S)$ is linearly independent.
2. S spans V if and only if $T(S)$ spans W .
3. S is a basis of V if and only if $T(S)$ is a basis of W .

What is the significance of isomorphisms? If two vector spaces V, W are isomorphic, $V \cong W$ (i.e., $\exists T : V \rightarrow W$ bijection), then V and W have the same linear algebraic properties (like dimension). We can think of V and W as the same space represented in two different ways.

However, identifying V with W usually requires a choice of an isomorphism $T : V \rightarrow W$. Usually there is no preferred (canonical) choice. It is therefore better, in general, not to identify different vector spaces that are isomorphic.

Definition 1.59: Rank

Let $T : V \rightarrow W$ be a linear map. We define the **rank** of T by

$$\text{rank}(T) := \dim \text{Im}(T)$$

Lemma 1.60: Rank of the Composition

Let $T : V \rightarrow W$, $S : W \rightarrow U$ be linear maps, where U, V, W are finite dimensional vector spaces. Then

1. $\text{rank}(S \circ T) \leq \min\{\text{rank}(S), \text{rank}(T)\}$.
2. If S is injective, then $\text{rank}(S \circ T) = \text{rank}(T)$.
3. If T is surjective, then $\text{rank}(S \circ T) = \text{rank}(S)$.

1.8 Writing Linear Maps using Bases/Coordinates

In this Section we'll assume all vector spaces to be finite dimensional unless otherwise stated. We'll write bases of a vector space V as an ordered list $\mathcal{B} = (v_1, \dots, v_n)$.

Definition 1.61: Coordinate Map

Let V be a vector space over K and $\mathcal{B} = (v_1, \dots, v_n)$ a basis of V . Let $v \in V$. We define the coordinate-vector of v w.r.t. the basis $\mathcal{B}$ as

$$[v]_{\mathcal{B}} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in K^n,$$

where $a_1, \dots, a_n \in K$ are the unique scalars in K s.t. $v = a_1 v_1 + \dots + a_n v_n$. Note that $[v]_{\mathcal{B}}$ is uniquely defined since every $v \in V$ has a unique representation as a linear combination of $v_1, \dots, v_n$.

Define a map $\Phi_{\mathcal{B}} : V \rightarrow K^n$ as

$$\Phi_{\mathcal{B}}(v) := [v]_{\mathcal{B}} \in K^n.$$

Proposition 1.62: $\Phi_{\mathcal{B}}$ is an Isomorphism

$\Phi_{\mathcal{B}}$ is an isomorphism

Remark 1.63. The inverse map to $\Phi_{\mathcal{B}}$ is given by

$$\Phi_{\mathcal{B}}^{-1} : K^n \rightarrow V, \quad \Phi_{\mathcal{B}}^{-1} \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} := a_1 v_1 + \dots + a_n v_n.$$

Corollary 1.64: Isomorphism Coordinate Space

Let V be a vector space over K with $\dim V = n \in \mathbb{Z}_{\geq 0}$. Then $V \cong K^n$.

Proof. Choose a basis $\mathcal{B}$ for V . Then $\Phi_{\mathcal{B}} : V \rightarrow K^n$ is an isomorphism. □

Remark 1.65. 1. The corollary does not hold for $\dim V = \infty$, at least not as stated.

2. We've seen that $V \cong K^n$, if V is finite dimensional, but in general there does **not** exist a canonical isomorphism $V \xrightarrow{\cong} K^n$. For example $\Phi_{\mathcal{B}}$ depends on the choice of basis $\mathcal{B}$.

Now we arrive at the main question. Let V, W be vector spaces over K , $n := \dim V, m := \dim W$, and let $T : V \rightarrow W$ be a linear map. Fix a basis $\mathcal{B}$ for V and a basis $\mathcal{C}$ for W . Consider the diagram

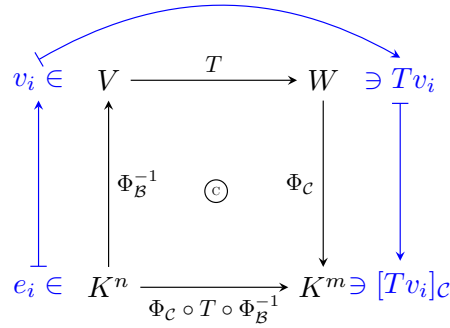
$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \Phi_{\mathcal{B}}^{-1} \uparrow & \text{\textcircled{C}} & \downarrow \Phi_{\mathcal{C}} \\ K^n & \xrightarrow{\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}} & K^m \end{array}$$

The 'C' means that the diagram is commutative.

Since Φ_B^{-1}, T, Φ_C are all linear maps, the map $\Phi_C \circ T \circ \Phi_B^{-1} : K^n \rightarrow K^m$ is also linear. By Lemma 1.47, any linear map $K^n \rightarrow K^m$ can be written using a matrix $A \in M_{m \times n}(K)$. In other words, there exists a unique matrix $A \in M_{m \times n}(K)$ s.t. $\Phi_C \circ T \circ \Phi_B^{-1} = T_A$. The matrix A is called the **representation matrix of T w.r.t. the bases $\mathcal{B}$ and $\mathcal{C}$** . We denote this matrix by $[T]_{\mathcal{C}}^{\mathcal{B}} \in M_{m \times n}(K)$.

1.8.1 Entries of the Representation Matrix

Let $T : V \rightarrow W$ be a linear map and $\mathcal{B} = (v_1, \dots, v_n)$ a basis for V , and $\mathcal{C} = (w_1, \dots, w_m)$ a basis for W . $T_A(e_i) = \Phi_C \circ T \circ \Phi_B^{-1}(e_i) = \Phi_C \circ T v_i = [T v_i]_{\mathcal{C}}$. But these are precisely the columns of A . In



other words

$$A = [T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [T v_1]_{\mathcal{C}} & \dots & [T v_n]_{\mathcal{C}} \\ | & & | \end{pmatrix} \in M_{m \times n}.$$

Another way to describe this matrix is: Let $A = (a_{i,j})$, where $a_{i,j}$ are defined uniquely by

$$T v_j = \sum_{i=1}^m a_{ij} w_i.$$

Proposition 1.66: Representation matrix

Let $T : V \rightarrow W$ be a linear map between two finite dimensional vector spaces V, W . Let $\mathcal{B}, \mathcal{C}$ be bases for V, W respectively. Then

$$\forall v \in V : [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [v]_{\mathcal{B}} = [T v]_{\mathcal{C}}.$$

1.8.2 Representing Composition of Linear Maps by Matrices

Let $T : V \rightarrow W, S : W \rightarrow U$ be linear maps. Let $\mathcal{A} = (v_1, \dots, v_n)$ be a basis for V , $\mathcal{B} = (w_1, \dots, w_m)$ a basis for W and $\mathcal{C} = (u_1, \dots, u_p)$ a basis for U .

Write $[T]_{\mathcal{B}}^{\mathcal{A}} = (a_{ij})$, $[S]_{\mathcal{C}}^{\mathcal{B}} = (b_{ij})$, $[S \circ T]_{\mathcal{C}}^{\mathcal{A}} = (c_{ij})$. By definition we have that

$$T(v_j) = \sum_{k=1}^m a_{kj} w_k \quad \forall 1 \leq j \leq n \quad (1.1)$$

$$T(w_j) = \sum_{i=1}^p b_{ij} u_i \quad \forall 1 \leq j \leq m \quad (1.2)$$

$$S \circ T(v_j) = \sum_{i=1}^p c_{ij} u_i \quad \forall 1 \leq j \leq n \quad (1.3)$$

But we can calculate $S \circ T(v_j)$ by using Equations (1.1), (1.2). Then we have

$$\begin{aligned} S \circ T(v_j) &= S\left(\sum_{k=1}^m a_{kj} w_k\right) = \sum_{k=1}^m a_{kj} S(w_k) = \sum_{k=1}^m a_{kj} \left(\sum_{i=1}^p b_{ik} u_i\right) \\ &= \sum_{k=1}^m \sum_{i=1}^p a_{kj} b_{ik} u_i = \sum_{i=1}^p \sum_{k=1}^m a_{kj} b_{ik} u_i = \sum_{i=1}^p \left(\sum_{k=1}^m b_{ik} a_{kj}\right) u_i \\ \implies c_{ij} &= \sum_{k=1}^m b_{ik} a_{kj}. \end{aligned}$$

1.9 Matrices

Definition 1.67: Matrix Multiplication

Let $A \in M_{m \times n}(K)$, $B \in M_{n \times p}(K)$. We define a new matrix $A \cdot B \in M_{m \times p}(K)$ which is called the product of A and B . Write $A = (a_{ij})$, $B = (b_{ij})$. The matrix $A \cdot B$ has entries (c_{ij}) , where

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Important: $A \cdot B$ is defined only when the number of columns of A equals the number of rows of B .

Conclusion: Let $T : V \rightarrow W$, $S : W \rightarrow U$ be linear maps between finite dimensional vector spaces. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be bases for V, W, U respectively. Then

$$[S \circ T]_{\mathcal{C}}^{\mathcal{A}} = [S]_{\mathcal{C}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{A}}.$$

Definition 1.68: Kronecker-Delta/Identity Matrix

Let $\mathcal{I}$ be an index set. Define for all $i, j \in \mathcal{I}$

$$\delta_{ij} := \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

This is called the **kronecker-delta**.

We define the **identity matrix** $I_n \in M_{n \times n}(K)$ as

$$I_n := \delta_{ij}, \quad I_n = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}.$$

Remark 1.69. Let V be a finite dimensional vector space and $n := \dim V$. Let $\text{id} : V \rightarrow V$ be the identity map. Let $\mathcal{B}$ be any basis for V . Then $[\text{id}]_{\mathcal{B}}^{\mathcal{B}} = I_n$. In other words, the matrix representation of the identity map w.r.t $\mathcal{B}$ & $\mathcal{B}$ is the identity matrix I_n .

Proposition 1.70: Associativity and Distributivity of Matrix Multiplication

$$1. \forall A \in M_{m \times n}(K), B \in M_{n \times p}(K), C \in M_{p \times q}(K) : (AB)C = A(BC) \in M_{m \times q}.$$

$$2. \forall A, B \in M_{m \times n}(K), C \in M_{n \times p}(K), C' \in M_{q \times m}(K) \text{ we have}$$

$$(A + B)C = AC + BC \in M_{m \times p}(K),$$

$$C'(A + B) = C'A + C'B \in M_{q \times n}(K).$$

$$3. \forall A \in M_{m \times n}(K) : I_m A = A I_n = A.$$

$$4. \forall \alpha \in K, A \in M_{m \times n}(K), B \in M_{n \times p}(K) : \alpha(AB) = (\alpha A)B = A(\alpha B).$$

Remark 1.71. Matrix multiplication is in general **not** commutative, i.e., $\exists A, B$ s.t. $AB \neq BA$.

In $M_{n \times n}(K)$ we can add/subtract matrices and also multiply matrices. There is also a unit I_n . These operations satisfy properties like associativity, distributivity, etc. From this we get that $M_{n \times n}(K)$ is a (non-commutative) **ring** with a unity.

Definition 1.72: Invertibility

An $n \times n$ matrix $A \in M_{n \times n}(K)$ is **invertible** if $\exists B \in M_{n \times n}(K)$ s.t. $AB = BA = I_n$.

Remark 1.73. If A is invertible, then there exists only one matrix B s.t. $AB = BA = I_n$.

Proof. Assume $AB = BA = I_n$ and $AC = CA = I_n$. Then

$$B = BI_n = B(AC) = (BA)C = I_n C = C.$$

□

Definition 1.74: The Inverse Matrix

If A is invertible, the unique matrix B s.t. $AB = BA = I_n$ is called **the inverse of A** and is denoted by A^{-1} . So, we have $AA^{-1} = A^{-1}A = I_n$.

We denote the set of all invertible $n \times n$ matrices by

$$GL_n(K) = GL(n, K) := \{A \in M_{n \times n}(K) \mid A \text{ is invertible}\}.$$

This set is called the general linear group.

Proposition 1.75: Structure of the General Linear Group

$GL_n(K)$, endowed with matrix multiplication $(A, B) \mapsto A \cdot B$ is a group. The unity is I_n . The inverse of $A \in GL_n(K)$ is A^{-1} . Moreover,

$$\forall A, B \in GL_n(K) : (A^{-1})^{-1} = A, \quad (AB)^{-1} = B^{-1}A^{-1}.$$

Corollary 1.76: Unique Solution for Linear System of Equations

Let $A \in GL_n(K)$. Then $\forall b \in K^n$ the linear system of equations $Ax = b$ has a unique solution, namely $x = A^{-1}b$.

Proposition 1.77: Invertibility and Isomorphisms

Let $A \in M_{n \times n}(K)$. Then $A \in GL_n(K)$ if and only if $T_A : K^n \rightarrow K^n$ is an isomorphism. Moreover, $(T_A)^{-1} = T_{A^{-1}}$.

Definition 1.78: Upper/Lower Triangular and Diagonal Matrices

A matrix $A \in M_{n \times n}(K)$, $A = (a_{ij})$, is called

- **Upper triangular** if $a_{ij} = 0 \quad \forall i > j$, i.e.,

$$\begin{pmatrix} * & \dots & \dots & * \\ 0 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & * \end{pmatrix}.$$

- **Lower Triangular** if $a_{ij} = 0 \quad \forall i < j$, i.e.,

$$\begin{pmatrix} * & 0 & \dots & 0 \\ \vdots & \ddots & & \vdots \\ \vdots & \ddots & \ddots & 0 \\ * & \dots & \dots & * \end{pmatrix}.$$

- **Diagonal** if $a_{ij} = 0 \quad \forall i \neq j$, i.e.,

$$\begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}.$$

Lemma 1.79: Multiplication of Triangular/Diagonal Matrices

Let A, B be two diagonal (respectively both upper triangular, respectively both lower triangular). Then $A \cdot B$ is of the same type.